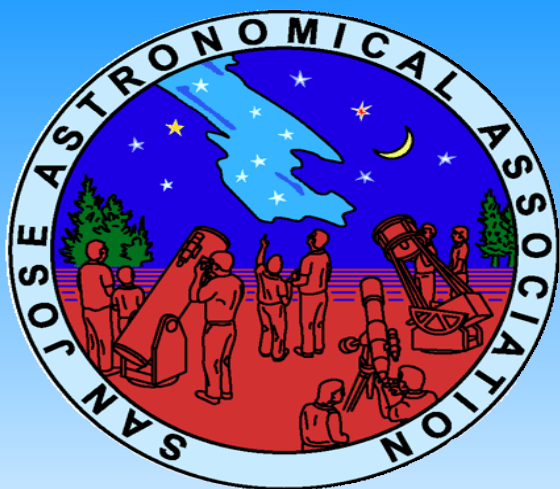




EPHEMERIS

June 2012

SJAA Activities Calendar

Jim Van Nuland

June

- 2 General Meeting. Board meeting (*) at 6:00; Social Time at 7:30; General Meeting at 8:00. Our speaker is Dr. Peter Nugent, speaking on supernova 2011fe.
- 3 Solar Program: H-alpha observing at Houge Park. Also, Telescope tune-up time (see page 5). Sun party hours: 2:00 until 4:00 p.m.
- 5 Transit of Venus — the planet can be seen crossing the face of the sun. Do NOT try this yourself; you will risk your eyesight! Sun&Venus Party at Houge Park, 2:30 p.m. to 6:00 p.m.
- 15 Astronomy Class at Houge Park. 8:45 p.m. The topic: Eyepieces, filters, finders and accessories. (outdoors)
- 15 Houge Park star party. Sunset 8:30 p.m, 10% moon rises 3:38 a.m. Star party hours: 9:45 until midnight.
- 16 Dark-Sky weekend. Sunset 8:30 p.m, 5% moon rises 4:20 a.m.
- 23 Dark-Sky weekend. Sunset 8:32 p.m, 20% moon sets 11:09 p.m. Henry Coe Park's "Astronomy" lot has been reserved.
- 29 Houge Park star party. Sunset 8:32

p.m, 82% moon sets 2:48 a.m. Star party hours: 9:45 until midnight. Includes "Introduction to Observational Astronomy" class for SJAA members.

July

- 1 Solar Program: H-alpha observing at Houge Park. Also, Telescope tune-up time. Sun party hours: 2:00 until 4:00 p.m.
- 7 General Meeting. Board meeting (*) at 6:00; Social Time at 7:30; General Meeting at 8:00. Our speaker is Dr. To B. Announced.
- 13 Astronomy Class at Houge Park. 9:15 p.m. The topic: Summer Constellations / Highlight Objects. (outdoor)
- 13 Houge Park star party. Sunset 8:29 p.m, 22% moon rises 2:17 a.m. Star party hours: 9:30 until midnight.
- 14 Dark-Sky weekend. Sunset 8:28 p.m, 15% moon rises 3:01 a.m.
- 21 Dark-Sky weekend. Sunset 8:24 p.m, 10% moon sets 9:44 p.m. Henry Coe Park's "Astronomy" lot has been reserved.
- 27 Houge Park star party. Sunset 8:19 p.m, 70% moon sets 1:34 a.m. Star party hours: 9:15 until midnight.

The Shallow Sky

Venus Transit

Akkana Peck

June 5 brings the last Venus transit until 2117, when Venus will pass across the face of the sun — the second of the only two Venus transits any of us are likely to see in our lives. (The first was in 2004, visible on the east coast of the US but not from here.)

Of course, this being a solar event, you need a safe solar filter to see it. No need for a fancy H-alpha filter — a white light filter is fine, preferably a glass filter over the aperture of the telescope.

(Don't use the kind that screw into the eyepiece! They can overheat and crack while you're looking through them.)

You don't need a big telescope, either: I used an Orion filter on my little 80mm f/7 refractor for the last Mercury transit and it worked great. And Venus is much larger than Mercury, so any telescope, or even binoculars, will show the transit. Venus's size is about 50 arcseconds (just under one arcminute), while the sun is about half a degree (30 arcminutes). Mercury only gets to 11 or 12 arcseconds, so if you've seen a Mercury transit, you can imagine how much easier and more spectacular a Venus transit can be.

Continued on page 2

24 hour news and information hotline:
(408) 559-1221
<http://www.sjaa.net>

If you use binoculars, either make sure that you have solar filters for both sides, or keep one side covered at all times.

If you absolutely can't find a solar filter in time for the transit, you can set up your telescope to project the sun's image onto a white board or sheet of paper. Use a cheap, low powered eyepiece for this: the eyepiece will get quite hot, and you don't want to risk damaging a fancy eyepiece. Be careful with solar projection — make sure nobody nearby can walk between the telescope and the surface you're using as a projection screen, or place their hands or eyes in the light path.

A web search for "solar projection" will uncover other tips.

When does this all happen?

Venus begins its ingress onto the disc of the sun on 3:06 on the afternoon of June 5. The transit continues until after the sun sets at 8:26. So we won't get to see egress. Venus's exit from the face of the sun, but it's the mirror image of what we'll see at ingress.

Ingress has two parts: first contact, when the edge of Venus's disk first touches the outside of the sun's disk, and "internal ingress" or second contact, when Venus's disk is fully inside that of the sun. Second contact is the most interesting period of the transit: it's when the famous "black drop effect" occurs, where the black circle of Venus can seem to elongate into a teardrop shape as it "tears away" from the edge of the sun. The black drop caused a lot of problems for early Venus transit observers, since it makes it difficult to record the exact second contact time. The effect seems to be caused by blurring from our own atmosphere (poor seeing) combined with telescope diffraction. So the steadier your seeing is, and the bigger and better your optics, the less likely you are to see the black drop.

If you have especially good seeing before second contact, try looking for the aureole, an arc of light just outside of the solar disk made by the refraction of sunlight through Venus's atmosphere.

Amazingly, the aureole has the same surface brightness as the sun's surface, and is said to be possible to see even through a solar filter. That's something you'll never see in a Mercury transit!

Here's the time table for San Jose (from the table on NASA's eclipse website):

First contact:	3:06:20 p.m.
Internal ingress:	3:23:56 p.m.
Maximum transit:	6:25:30 p.m.
Sunset:	8:26 p.m.

At first contact, the sun will still be high, 60 degrees up. By maximum transit the pair will have sunk to 21 degrees, still plenty high enough to see the spectacle. Photographers will want to wait around for sunset for a chance at some spectacular photos, such as this sunrise photo (see above) taken by SJAA member Bill Arnett from Chicago during the 2004 Venus transit.

Want more details or times from other locations?

Go to transitofvenus.org for the full scoop.

What about other planets to watch this month? Who cares? Okay, maybe you do care. So here's the list:

Saturn rules the evening sky, transiting just before midnight.

Mars, too, hangs in the evening sky and is visible until midnight, though it's



small, distant and not very high.

Neptune doesn't rise until around midnight.

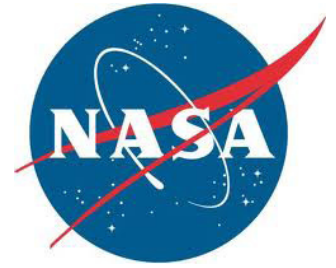
Mercury moves into the early evening after the first week of June.

Jupiter moves into the dawn sky, where it joins Uranus.

Pluto, at opposition on June 29, is visible all night ... but it's in the heart of the Sagittarius Milky Way next to M25, so you'll have a tough job finding it. But you'll see some great scenery along the way!

Golden State Star Party

GSSP is July 18-22. For more info: <http://www.goldenstatestarparty.org>



Thank Goodness for Magnetism

Dr. Tony Phillips

Only 93 million miles from Earth, a certain G-type star is beginning to act up.

Every 11 years or so, the solar cycle brings a period of high solar activity. Giant islands of magnetism—"sunspots"—break through the stellar surface in increasing numbers. Sometimes they erupt like a billion atomic bombs going off at once, producing intense flares of X-rays and UV radiation, and hurling massive clouds of plasma toward Earth.

This is happening right now. Only a few years ago the Sun was in a state of deep quiet, but as 2012 unfolds, the pendulum is swinging. Strong flares are becoming commonplace as sunspots once again pepper the solar disk. Fortunately, Earth is defended from solar storms by a strong, global magnetic field.

In March 2012, those defenses were tested.

At the very beginning of the month, a remarkable sunspot appeared on the Sun's eastern limb. AR1429, as experts called it, was an angry-looking region almost as wide as the planet Jupiter. Almost as soon as it appeared, it began to erupt. During the period March 2nd to 15th, it rotated across the solar disk and fired off more than 50 flares. Three of those eruptions were X-class flares, the most powerful kind.

As the eruptions continued almost non-stop, Earth's magnetic field was

buffeted by coronal mass ejections or "CMEs." One of those clouds hit Earth's magnetosphere so hard, our planet's magnetic field was sharply compressed, leaving geosynchronous satellites on the outside looking in. For a while, the spacecraft were directly exposed to solar wind plasma.

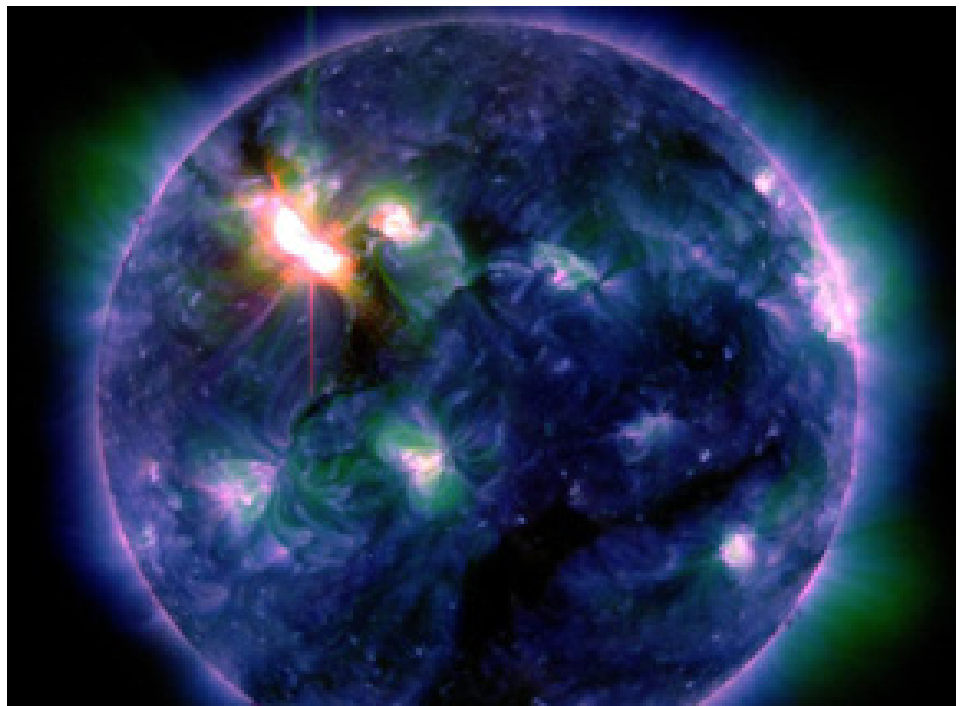
Charged particles propelled by the blasts swirled around Earth, producing the strongest radiation storm in almost 10 years. When those particles rained down on the upper atmosphere, they

dumped enough energy in three days alone (March 7-10) to power every residence in New York City for two years. Bright auroras circled both poles, and Northern Lights spilled across the Canadian border into the lower 48 states. Luminous sheets of red and green were sighted as far south as Nebraska.

When all was said and done, the defenses held—no harm done.

This wasn't the strongest solar storm

Continued on page 4



Multiple-wavelength view of X5.4 solar flare on March 6, captured by the Solar Dynamics Observatory (SDO) in multiple wavelengths (94, 193, 335 angstroms). Credit: NASA/SDO/AIA

in recorded history—not by a long shot. That distinction goes to the Carrington Event of September 1859 when geomagnetic activity set telegraph offices on fire and sparked auroras over Mexico, Florida, and Tahiti. Even with that in mind, however, March 2012 was remarkable

It makes you wonder, what if? What if Earth didn't have a magnetic field to fend off CMEs and deflect the most energetic particles from the Sun.

The answer might lie on Mars. The red planet has no global magnetic field and as a result its atmosphere has been stripped away over time by CMEs and other gusts of solar wind. At least that's what many researchers believe. Today, Mars is a desiccated and apparently lifeless wasteland.

Only 93 million miles from Earth, a G-type star is acting up. Thank goodness for magnetism.

With your inner and outer children, read, watch, and listen in to "Super Star Meets the Plucky Planet," a rhyming and animated conversation between the Sun and Earth, at <http://spaceplace.nasa.gov/story-superstar>.

This article was provided by the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration.



AUCTION

2012



From top to bottom on the left: Rick Neuschaefer and Rob Jaworski look on as Mark Wagner and Greg Claytor fire up the computer for the start of the auction; several Dobsonians were for sale; a number of cameras were made available.



**From top to bottom starting above:
a large Meade LX75 refractor looks
tempting; Lee Hoglan is happy that
some of the loaner scopes will be sold
allowing for a refresh of the telescope
loaner program that he leads; Mark
Wagner and Greg Claytor are feeling
very positive about the auction; a
number of folks came out for the
auction; Kevin Medlock did another
fantastic job as auctioneer. All photos
from the auction by Ed Wong. We
thank him for his efforts.**



The Last Month In Astronomy

MAY-09-2012 **Vesta News**

NASA's Dawn spacecraft is showing that Vesta is more like the Earth's moon than a typical asteroid. It appears to have an iron core. In fact, it probably had a magma ocean which means it underwent differentiation where distinct layers are created. This confirms that a particular group of meteorites found on Earth originated on Vesta, probably 6% of all meteorites that fall to Earth. <http://www.jpl.nasa.gov/news/news.cfm?release=2012-132>

MAY-08-2012

Spitzer "sees" Super Earth

The Spitzer infrared observatory has directly measured light from a Super-Earth sized exoplanet. Usually it is impossible to see light from an exoplanet because it is lost in the glare of its star. But in the infrared the ratio is not quite as bad. <http://www.jpl.nasa.gov/news/news.cfm?release=2012-127>

MAY-03-2012

Opportunity still rocking

The Mars Rover Opportunity, already 8 years beyond its expiration date, has found "the strongest evidence for water that we've ever seen with Opportunity" according the PI Steve Squyres of Cornell. The other rover, Spirit, stopped communicating in 2010. The next rover, Curiosity, should land on Mars on August 5. <http://www.jpl.nasa.gov/news/news.cfm?release=2012-124>



APR-30-2012

Pre-planetary Nebula

By now you know that when astronomers talk about planetary nebulae, they don't mean anything related to planets. But pre-planetary nebulae are different. One such nebula is the Egg Nebula. Hubble recently imaged this object (see image on the left). The central star cannot be seen, in fact, it might be a binary star system. The nebula is 3000 light-years away although that distance has large error bars. The image was taken with Hubble's Wide Field Camera 3. Credit: ESA/Hubble, NASA <http://www.astronomy.com/News-Observing/News/2012/04/Hubble%20images%20searchlight%20beams%20from%20a%20preplanetary%20nebula.aspx>

APR-26-2012

Peter J. does it again

Peter Jenniskens had uncovered some meteorites fragments after a large meteor was seen over California. These fragments were found in the Gold Country. Later he and Mike Koop found more fragments from the air. The original meteor is estimated to weigh more than 154,000 pounds. An event of this size occurs about once a year. This event occurred on April 22. <http://www.sfgate.com/cgi-bin/article.cgi?f=/c/a/2012/04/25/BAOV1O9B4H.DTL&tsp=1> ; http://www.msnbc.msn.com/id/47286726/ns/technology_and_science-space/t/scientists-take-skies-airship-look-meteorites/

APR-25-2012

Earth under attack

A pair of studies published in Nature gives evidence that the heavy bombardment of Earth lasted much longer than previously believed. In fact it lasted for almost the entire first half of Earth's history. Earlier computer simulations suggested the Late Heavy Bombardment (LHB) lasted from 4.1 billion to 3.8 billion years ago. Calling this into question is the large number of large lunar craters that are much younger than 3.8 billion years old. Instead, the asteroid belt, currently starting 2.1 AU from the Sun, might have started just 1.7 AU up until 2 billion years ago. This would cause many more objects to become dislodged from the belt and come barreling into Earth. To bolster this idea another study looked for the results of large impacts on Earth. Brandon Johnson from Purdue University says "Our work shows that a lot more big asteroids — meaning dinosaur-killer or larger — were hitting Earth well after the current idea of when it ended." <http://www.nature.com/news/ancient-asteroids-kept-on-coming-1.10504>

APR-24-2012

Galaxy Schizo

It seems the galaxy NGC4594, aka The Sombrero Galaxy, has two personalities. In visible light it appears to be a fairly typical spiral galaxy with a rather bright halo. But in infrared light (thank you Spitzer satellite observatory) the galaxy appears to be an elliptical galaxy. "The Sombrero is more complex than previously thought," according to Dimitri Gadotti of the ESO in Chile. "The only way to understand all we know about this galaxy is to think of it as two galaxies, one inside the other." <http://www.jpl.nasa.gov/news/news.cfm?release=2012-115>



SETIcon, envisioned and organized by the SETI Institute, is a unique, entertaining, and enlightening public event where science and imagination meet. SETIcon brings together innovative scientists, science fiction authors, space and science artists, space lovers, and the curious and adventurous everywhere for a 3-day public celebration and exploration of space, real science, science fiction, technology, imagination, and science education.

SETIcon II will land at the Hyatt Regency - Santa Clara, California - June 22-24, 2012

There is no other event in the world like SETIcon that explores space and the human imagination through the lens of real science, attracting global interest and participation. This is not a science conference with technical lectures (SETI Institute scientists lecture all over the world), nor Comic-Con or other Star Trek type conventions. **SETIcon creates a new channel of discussion where real science and imagination meet.**

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Get your tickets now and get in on the ground floor of what will surely be a memorable and entertaining event! Visit www.seticon.com for a detailed list of speakers, other important information, and to purchase tickets. **Register now for best prices!**

We look forward to seeing you this year at SETIcon II. It will be out of this world!

Sincerely,
The SETIcon II Event Team

For more information on SETIcon II and to buy tickets: www.seticon.com
Questions?: seticon@seti.org

It Must Be Astronomical ...

Directions to Houge Park

Houge (rhymes with "Yogi") Park is in San Jose, near Campbell and Los Gatos. From Hwy. 17, take the Camden Avenue exit. Go east 0.4 miles, and turn right at the light, onto Bascom Avenue. At the next light, turn left onto Woodard Road. At the first stop sign, turn right onto Twilight Drive. Go three blocks, cross Sunrise Drive, then turn left into the park.

From Hwy. 85, take the Bascom Avenue exit. Go north, and turn right at the first traffic light, onto White Oaks Road. At the first stop sign, turn left onto Twilight Drive. You will now be passing the park. Turn right at the first driveway, into the parking lot.

Loaners

The telescope loaner program has been revamped. The program now includes QuickSTART, program geared to those new to astronomy. Please check it out at <http://www.sjaa.net/loaners.shtml>.

School Star Parties

Completed Events

	Total Sched.	Good Sky	Partial Success	Cloudy Fail	Cancel at noon
Jul	0				
Aug	1	1			
Sep	1	1			
Oct	6	3			3
Nov	13	9		1	3
Dec	4	4			
Jan	4	0	1		3
Feb	10	6		1	3
Mar	10	1	3		6
Apr	5	3	1		1
Total	54	28	5	2	19
Scheduled					
May	1				
Jun	1				
Jul/Aug	2				
Total	10				

As of May 1, 2012

School Star Party Link

For information on school star parties including how to schedule one see <http://www.sjaa.net/school.shtml>.

Officers and Board of Directors

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VP Greg Claytor
Sec Rob Jaworski
Tres Robert Armstrong
Dir Lee Hoglan
Dir Rich Neuschaefer
Dir Rod Norden
Dir Kevin Roberts
Dir Michael Packer

School Star Party Chairman

Jim Van Nuland (408) 371-1307

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 School Star Parties schools@sjaa.net
 Ephemeris ephemeris@sjaa.net

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Members Email Lists:
<http://www.sjaa.net/majordomo.html>

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San Jose Astronomical Association
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San Jose Astronomical Association Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

☐ **New** ☐ **Renewal** (Name only if no corrections)

Membership Type:

- ☐ Regular — \$20
☐ Regular with Sky & Telescope — \$53
☐ Junior (under 18) — \$10
☐ Junior with Sky & Telescope — \$43

Subscribing to Sky & Telescope magazine through the SJAA saves you \$5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2 months lead time.)

☐ I prefer to get the Ephemeris newsletter in print form (Add \$10 to the dues listed on the left). The newsletter is always available online at <http://ephemeris.sjaa.net>
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Bring this form to any SJAA Meeting or send to the club address (above). Please make checks payable to "SJAA".

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